The irruption theory: How mind and matter meet along the route to consciousness

An interview with
Tom Froese

By Matthieu Koroma

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Abstract

In this interview, Tom Froese and Matthieu Koroma address the irruption theory, an account of how mental actions manifest themselves as unexplained variability in brain and bodily activity. We discuss how this approach can be applied to trace back consciousness in (neuro)physiological signatures such as scale free activity and informational complexity. We highlight the study of altered states of consciousness as an opportunity to disentangle the physiological correlates of consciousness from agency. We conclude by addressing how irruption theory provides insights into the therapeutic effects of psychedelics and the importance of crafting an appropriate context to consciously shape agency and behavior.

Keywords: *irruption theory, agency, free-will, volition, consciousness, psychedelics*

With your team, the Embodied Cognitive Science Unit at the Okinawa Institute of Science and Technology Graduate University (OIST, Japan), you develop an approach of the mind drawing on theories and methodologies from philosophy, physical science, computational modeling, neuroscience and psychology. What is the importance for you of interdisciplinarity in the study of the mind and how is it enacted in the research practice of your team?

We are interested in understanding the human mind in all its complexity, which basically means starting with the recognition that people are physical, biological, psychological, social, cultural, and of course also conscious beings. The human being integrates all of these domains of description, and so our aim is to do justice to this inherent complexity. The very nature of the target phenomenon thereby forces us to find a way of doing science that can put these usually distinct disciplines into a productive dialogue with each other. In practice, this means keeping an open mind, respecting the specificities characterizing different domains of investigation, and finding a way to work with and capture the resulting diversity, and even irreducible ambiguity, without aiming to reduce them to a simple unity. Ambiguity is a positive characteristic of being human, rather than a shortcoming of the interdisciplinary method.

You recently introduced the irruption theory as a principled account of motivated activity and consciousness (Froese, 2023, 2024). Could you explain this theory and how it relates to other theories of the mind and consciousness?

The irruption theory asks the question: *how does mind matter?* How do I know that certain aspects from someone's mental and conscious activity make a difference in the material world? From the perspective of the natural sciences, the first-person point of view of consciousness is an unobservable, rather than an observable, because we cannot directly observe it when we look at the level of description of the brain and the body. Natural sciences have moved over their evolution beyond direct observables already ages ago and deal with non-observables all the time. We have lots of methods for working with unobservables, yet most of cognitive neuroscience for some reason is obsessed about finding observables, matching consciousness with neural events that are supposed to be the basis of the experience, the neural correlates of consciousness.

> "How do I know that certain aspects from someone's mental and conscious activity make a difference in the material world?"

The route to consciousness is still very open but the general idea is the following: while we cannot directly measure the subjective level from brain signals in the usual scientific way, we can still have the ambition to measure how the subjective mind makes a difference in the body described as a physical system and vice-versa. Our approach is a bit like predictive processing where surprise — or prediction error — in brain signals is reflected as a difference between bottom-up and top-down processes rather than only the consequence of one or the other. In that sense, the irruption theory is realist about the physical world making a difference to consciousness and vice-versa. It is neither an identity or physicalist theory that would say that consciousness and the physical body are the same thing nor an idealist or epiphenomenalist theory that would treat the physical world and consciousness as separate. It is a "not one not two kind" of approach like Varela used to say.

> "We can measure how the subjective mind makes a difference in the body described as a physical system and vice-versa."

What could distinguish consciousness from other natural processes?

A kind of self-reference that is suggested by certain kinds of material configurations, like autopoietic, adaptive systems, may be the source of an internal perspective on the world from where the subjective point of view could come from. There is also the question whether the origin of these kinds of systems, the origins of life, is also the origin of everything that is qualitative in the universe. If you really want to be realist and non-reductionist about something like "goodness" in a physical system, you have to show that another, purely physical or dynamical account cannot do full justice to the phenomenon described as being good. If there is such an alternative account, then there is no need to appeal to what is good for the system to explain what is happening. And it doesn't help to posit representational content either, because that would simply presuppose what we are trying to explain.

To do better, we need a notion of what it means for a goal or value to make a difference to the system in its own right: we call that a participation criterion (Froese et al., 2023). The base phenomenon is a primitive one: a “doing” rather than a “happening” with normative conditions of success or failure that are making a difference to something. For example, if Earth fell into the Sun because of a fluke in gravity, this would not be a failure of earth’s trajectory because the planet as such is not actively regulating its distance to the sun. Irruption theory is an answer to this challenge of differentiating a “doing” from a mere “happening” (Froese, 2023). The first irruption theory paper was a little bit simplistic, as I only looked in one direction, which was characterizing intelligent behavior and agency, whereas the emergence of conscious experiences actually points in the other direction, which leaves a trace I call absorption (Froese, 2024).

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What would a signature of consciousness in the material world look like according to the irruption theory?

If we are realist about the mind and we accept a non-reductionist take, we can measure how much being a conscious agent, one engaged in volitional action, actually makes a difference in the neural signal and vice-versa. If we use an objective method to study quantitatively what happens in your subjective mind makes a difference in the world, there will be an unexplained variability in neural signals that is not reducible to the description of the physical system (Froese, 2023). The world-to-mind direction of influence, that is absorption and not irruption, would be a hidden variable, a kind of counterfactual condition, that shows as a decrease rather than increase in variability in the description of the physical system (Froese, 2024). The question is then what kind of neural signals reflect the phenomenon of becoming conscious of the world?

> “If what happens in your subjective mind makes a difference, there will be an unexplained variability in neural signals that is not reducible to the description of the physical system.”

An example of such a signal that we are investigating is what I like to refer to as “counterfactual chirality.” Chirality refers to situations like handedness with pairs that are symmetric but not identical. If I put my right and left hand on top of each other, they do not match; they are related but not the same. Asymmetry is all over the place in all kinds of biological systems and even in neurodynamics and we are trying to develop a method to test if such symmetry and anti-symmetry in the neural phase space could be a signature for consciousness.

Investigating the dimensionality of signals could be a way of operationalizing this question. One may have a perceptual discrimination task for which people respond whether they saw an image or not. From there, you can investigate the dimensionality of the neural responses. An increase in the dimensionality of the neural signals could be related to the fact that, for example, an image consciously seen triggers memories of a person suddenly opens a whole mental world to the extent that some of that mental world shapes the neural response, which is something that cannot directly be inferred from neural signals.

Some people in my group keep pushing me to say that structured noise, like pink noise, may have long-term dependencies in the signal over time that could well fit a marker of consciousness. Pink noise is already part of the general scientific discussion of the signatures of life and mind: it's a noise signal but the noise is in itself structured temporally in a way that is not independent from what's happened in the past (Van Orden et al., 2005, 2011; Ward & Greenwood, 2007). This seems like a quality that can be studied carefully to investigate whether temporal dependencies possibly reflecting self-reference are associated with the temporal structure of consciousness.

Signatures of scale freeness should, however in my opinion, not be enough to characterize the trace of consciousness. There are many lifelike phenomena in the universe that have self-organizing properties and are adaptively robust. For example, in Japan, we have frequent earthquakes and it is easily observed that earthquakes follow a power law. Small ones are more frequent than large ones, and this relationship can be observed in the slope of the frequency spectrum of seismographs. I don't think that it is because there's consciousness at play deep in the earth, and yet, we observe long range temporal dependency so these properties may be necessary but not sufficient to talk about the efficacy of mind. We must remain careful with identifying consciousness with this and other physical properties.

Bjorn Brembs is a researcher at Freie Universität Berlin that works on free will in flies (Brembs, 2011). One of his experiments is putting a group of flies in a bidirectional box. 80% of them go in one direction while 20% go in the other direction. The 20% that went in one direction and the 80% that go in the other direction are then put in the same box and the experiment is done again. Here again, there is an equal split of 20% and 80%. His argument is that there's a kind of inner noise generator that is producing this 80-20% divide that can be a basis of spontaneous behavior generation. How could the irruption theory characterize such a link between noise generation and spontaneous behavior?

It makes a lot of sense to me to investigate the basis of voluntary processes in insects in terms of such stochastic phenomena. We could go further, even to single cell organisms for which their whole genome and biochemical pathways are mapped. Even at that level, we get surprises such as transcriptional burstiness phenomena in real time measurements of gene expression levels that is a headache for engineering of chemical productions purposes because we would rather have a very stable level of expression. Gene expression is actually stochastic and even in these systems, we can ask what is it about regulation that needs this intrinsic noise aspect to it. With these stochastic processes, there are degrees of freedom that sometimes get tapped and sometimes don't. When they get tapped, they look like stochastic events that are not predetermined.

Could we be able to capture moments when consciousness and the mind would make a significant difference in bodily signals and neural markers of consciousness? How would they then be conceived in the irruption theory framework?

I think we underestimate how intelligent our body actually is without our conscious control. Most of what we consider intelligent goal-directed behaviors is actually unconscious. I don't fall out of this chair because my body keeps me in balance and this remains outside the scope of my awareness. Slightly separating mind and body also gives the body its due. The role for conscious intervention is relatively minor here and more about context switching, for example should I shift my pose, switch conversational topics, get to the next meeting room: those transition moments are probably where most of the conscious mind is required in terms of intervening in the existing bodily trajectories.

Operationalizing these processes in terms of neural signals is hard because systems are complex and there are always multiple possible explanations to account for them. A way of thinking about the literature which resonates with some of the topics we're discussing is the readiness potential, a phenomenon uncovered by Libet describing a ramping up noise term preceding conscious voluntary decision (Libet, 1985).

I like the work in this area of Aaron Schurger on the stochastic accumulator model. The basic idea is that neural activity stochastically fluctuates and eventually crosses a threshold that triggers a decision such as moving a finger when the fluctuations are large enough (Schurger et al., 2012). The philosophical challenge is to provide a story of why that noise should be associated with agency and/or free will (Schurger et al., 2021). Irruption theory provides a broader theoretical framework giving us *a priori* reasons for looking for these kinds of signals that are already there in the data and to explore how they can relate to agency and consciousness.

Can the study of altered states of consciousness be informative about the dissociation between agentive control and consciousness and their relation to markers of consciousness?

A lot of inspiration for the irruption theory comes from what happens in the brain during psychedelic states, where there is an increase of neural information metrics such as entropy and complexity (Carhart-Harris et al., 2014; Schartner et al., 2017). The question whether this rise in neural complexity reflects awareness or increased agency is difficult to disentangle in psychedelics because a rise in awareness in psychedelics also leads to a more challenging experience and thus usually requires more cognitive effort. There's a couple of studies on people undergoing administration of anesthesia who, while slowly falling unconscious, are still able to engage in a cognitive task (Boncompte et al., 2021). It turns out that neural complexity goes down for people falling unconscious and stopping to work on the task properly. But neural complexity is even higher than during wakeful baseline for people who hang on, despite the fact that they're almost falling unconscious. For me, it suggests that maybe irruption theories are on the right track because they are associating complexity with mental or cognitive effort rather than level of awareness *per se*.

Can this interplay between agency and consciousness as conceptualized in the irruption theory provide insight into the application of psychedelics for psychiatric diseases?

If the psychedelic state is characterized by increased neural entropy, it could explain why they're effective at treating certain kinds of conditions where agency seems to be impaired (Hipólito et al., 2023). What the psychedelics do is loosen up degrees of freedom in the system such that something like irruption can happen more easily, meaning that the intervention of the mind in the bodily basis of behavior generation is facilitated. That would be one way of thinking about that, but more work certainly needs to be done.

Note that it's not just about increasing neural entropy, because too much openness could be counterproductive in the long run. The traditional ritual way of managing these kinds of altered experiences is by creating special occasions that are social events most of the time basically preparing the body and the mind to enter that space and reintegrating them during and after the process would be very important to consider for mental health.

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As a concluding remark, I feel that we often forget from a first-person perspective to take into account the context when we think about our own actions. While being conscious allows us to be sensitive to many factors from the context and deciding between possible actions, a lot about our own ability to direct actions is actually driven by constraints that come from the world.

Indeed, we cannot be in charge of everything that we do and happen to us and it's okay, it is just the nature of things, and we shouldn't beat ourselves up if we sometimes do the wrong things. We can't directly be in charge of everything happening in the "here and now" and which affordances of the environment we may follow. There is an irreducible gap between our intention and our behavior; we can open a space of possibilities, but the precise action that emerges is self-organized on the basis of a history of organism-environment interaction. Where our responsibility lies is to create the right conditions along the path: that is a long-term project; it's about the cultivation of habits, about skills, about selecting the right environments, because that's really where the creation of the conditions of behavior generation takes place.

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Consciousness :

from molecular to social scales.

An interview with

Jean-Pierre Changeux

By Guillaume Dumas

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Abstract

In this interview, Guillaume Dumas reaches out to Jean-Pierre Changeux to offer a wide-ranging view on consciousness, spanning over its fundamental molecular mechanisms up to its social dimension. They further discuss the exploration of altered states of consciousness and its relation to artistic practice. They finally tackle the opportunities of some of the prominent fields of future research such as artificial intelligence and highlight the importance of multidisciplinary research in advancing the study of consciousness.

Keywords: *consciousness, neuromodulation, global neuronal workspace, AI, art, psychedelics*

Your work has significantly influenced our understanding of the neural underpinnings of consciousness. How do you view the evolution of consciousness studies since your pioneering forays into the field?

First of all, in the past decades, an authentic scientific approach to conscious processing was initiated by Francis Crick with his "spotlight of attention" hypothesis about visual awareness. His provocative position paradoxically gave legitimacy to the reintegration of the word consciousness into contemporary scientific language but also offered the opportunity to develop a theoretical and experimental understanding of consciousness.

From the start, I would like to strongly underline the importance of theory, which is not often the case, in those days, in biological and physiological sciences, at variance with physics, where theory is dominant. However, in our discipline, theory without experimental tests does not make sense. Fortunately, in the field of consciousness research, new experimental tools became available, in the past decades, in brain imaging, genetics, molecular biology, chemistry, and AI… which — together with the joint use of experimental psychology — made possible a fair evaluation of theoretical hypotheses about consciousness right away. An "integrative" strategy based on the embodiment of cognitive functions, establishing a causal relationship between neuro-molecular processes, modeling, and conscious processing, became accessible.

I dislike the word "neural correlates" of consciousness but prefer "neural mechanisms" or, better "neuro-molecular mechanisms" of consciousness. The future is there, together with a particular emphasis on the strong relationships established by the human brain with its multiple environments biological, physical, social, and cultural. Concerning consciousness itself, up to now, it has been investigated almost exclusively with the individual brain and at its simplest levels. A very promising future is opened, with a similar "adapted" strategy, for the social brain and the higher levels of consciousness including theory of mind, rationality, language… A fascinating avenue!

Throughout your career, what discoveries or insights have most altered your own understanding of consciousness?

I would prefer not to use "altered" but "progress" in my understanding of consciousness. In my last book ("*Le Beau et la Splendeur du Vrai*") I have described my paths from allostery, "Neuronal man", up to the Global Neuronal Workspace. Early on, in Neuronal Man, I had a section (Ch 5.9) on the "substance of the spirit" where I introduced the idea that a "surveillance system" composed of very divergent neurons and their reentries, such that a "spider web" (of connections), may form and function as a "whole" and as a consequence consciousness would "emerge."

The teaching at the Collège de France gave me the opportunity to further reflect on the fascinating history of the concepts relative to consciousness in the course of the past centuries, starting from Lamarck's "*sentiment interieur*," Hughling Jackson's hierarchical organization, William James's stream of consciousness, Llinas's electrophysiology of conscious states and last but not least Baars's book "a cognitive theory of consciousness" (1988). In this book, Baars introduces the Global Workspace theory (GW), yet with little, if any, solid neurobiological foundations and no reference to the molecular level.

Beforehand, together with Stanislas Dehaene, I had elaborated neuro-molecular models of low-level cognitive functions such as the Wisconsin Card Sorting Task (Dehaene & Changeux 1991) and after reading Baars's, I decided to extend the same approach to the neurobiological and molecular mechanisms involved in the GW. We proposed, among other features, the contribution of a brain scale network of cortical neurons with long-range axons, spontaneous activity of the network, dopamine reward... as basic components of a Global "*Neuronal*" Workspace (GNW) (Dehaene, Kerszberg, Changeux 1998; Dehaene & Changeux, 2011). This was the beginning of an amazing story that is still quite alive!

In your opinion, how do psychoactive substances modulate neuronal networks, and what implications does this have for our understanding of consciousness?

Drugs modulate neuronal networks at the level of circuits of neurons releasing a well-defined neurotransmitter. They are intrinsic components of the circuits engaged in any defined brain functions. For example, dopamine is a neurotransmitter well known to play a fundamental role in reward, and cocaine/amphetamines block dopamine re-uptake by its presynaptic transporter. The issue is then: would some of these neuro-molecular circuits contribute to conscious processing? where and how? The original version of the Global Neuronal Workspace of 1998 was designed to account for the "Stroop task" and the reward was an essential part of the global network (as recently emphasized by Volzhenin et al., 2022).

Tasks subsequently used to test the GNW, like the "masking tasks", did not need explicit reward. Interestingly, the modeling of a masking task (Dehaene et al., 2003) led to the proposal of a contribution to two critical components of brain circuitry: the NMDA vs AMPA glutamate receptors. These would intervene differentially in top-down vs bottom-up "resonant" circuits engaged in conscious processing. Further modeling and experimental work confirmed that. Sooner or later, the fair understanding of conscious processing shall obligatorily include the molecular level!

Could you elaborate on the neurobiological mechanisms that you believe are most crucial in altering states of consciousness, particularly in relation to your research on nicotinic receptors and allosteric modulation?

Most often, the current studies of the states of consciousness in well-doing subjects rely upon behavior, clinics, or computational modeling. "Disorders" of consciousness like coma, vegetative state, or minimally conscious states… reveal the importance of critical elements of brain anatomy in the access to consciousness. What is usually reported as "altered" states of consciousness like, for instance, the hallucinations produced by psychotropic drugs, are especially interesting for me since they selectively target the *molecular level* on top of which the whole brain is built. There is a fundamental bottom-up top-down relationship between the molecular and the conscious level, which is frequently underestimated in the field.

A wide variety of pharmacological agents, natural or synthetic, are known to act differentially on brain molecules such as neurotransmitter receptors but also enzymes, transporters. Many of them bind directly to the "active" site and have the same or a more powerful effect as the neurotransmitter — they are referred to as agonists. Others block the physiological ligand at the same site in a mutually exclusive manner — they are referred to as competitive antagonists. Nicotine at high doses is used by Amazonian populations to "shape dreams" and by Australian Aboriginal people to cause trance-like states and "excite their courage in warfare". It is a potent agonist of the nicotinic acetylcholine receptor.

> “There is a fundamental bottom-up top-down relationship between the molecular and the conscious level, which is frequently underestimated in the field.”

The beloved Martin Fortier (from the Ecole Normale Supérieure) distinguished two main categories of hallucinogenic drugs: first, the serotonergic drugs like mushroom psilocybin, whose action resembles that of LSD which behaves as an agonist of the serotonin 5-HT_{2A} receptors, and second, the muscarinic drugs, which are competitive antagonists of the G-protein linked muscarinic receptor for acetylcholine.

In relation to our research on nicotinic receptors, it was discovered that some drugs do not act at the level of the receptor binding site nor at the level of the biologically active site (channel, G-protein site…) but target different *allosteric* modulatory sites distributed at different levels on the receptor molecule. One example is the benzodiazepines with potent anxiolytic properties which act as positive allosteric modulators of the $GABA_A$ receptor. Another example is related to the effects of hashish: tetrahydrocannabinol and cannabinoid ligands act as positive allosteric modulators of glycine receptors. About 80 allosteric modulators are used in the clinic (Changeux & Christopoulos, 2016)! A new allosteric pharmacology is born to investigate the disorders of consciousness.

Considering your extensive research in neuropharmacology, what ethical considerations should guide us in exploring altered states of consciousness through substances?

This is a necessary but challenging question for me as the former Chair of the French National Bioethics Commission. My first reaction is that because of the diversity of drugs and their mode of action, each case should be examined and discussed by itself. The second reaction is that the subject’s informed consent should be received with a fair explanation of the experiment’s aim, which is most often difficult to really explain to the patient. The third is that there exists in many countries special committees for the protection of persons that have to be consulted.

Now, there is a particular issue about experimentation with drugs since many of them are addictive – there is a loss of control taking place where use leads to abuse – and, even more, some of these drugs at a single dose may have irremediable effects on the brain (*i.e.*, MPTP and Parkinson). Should it mean that the use of any of these drugs has to be forbidden? First of all, there is the use of drugs by shamans and native populations practicing shamanism. My view is that this traditional use of drugs is well controlled and, for millennia, was not shown to create irremediable alterations of consciousness. It might be accepted under well-defined social conditions.

The dangers of uncontrolled use of drugs are well established (tobacco, alcohol, cocaine, heroin, etc.), and, by all means, the general population should be protected against it. The severity of the constraints imposed should be dictated by the dangerousness of the drug. This becomes particularly important when some of these dangerous drugs are planned to be used as medicaments against well-defined brain diseases. Classical, among many examples, is the use of cannabinoids on intractable pain, the use of nicotine against Gilles de la Tourette disease or Ritalin against ADHD despite its close relationships with cocaine. There is a dual role of drugs which should be examined seriously.

A scientific debate about the clinical effect of the considered drug should take place to identify the positive effects together with the negative ones. An eventual prescription of any of such drugs should thus follow a prudence rule: the evaluation of the balance benefit-risks in the treatment of the disease. Moreover, each individual case has its own history and should be dealt with case by case.

In light of your collaboration with artists and your work on the neural mechanisms of aesthetics, how do you believe altered states of consciousness, whether induced by psychoactive substances or other means, influence the cognitive and neural processes underlying the creation and perception of art? What implications might these altered states have for our understanding of aesthetic experience and creativity from a neuroscientific perspective?

A number of important artists were taking drugs in their creative process. Famous examples are Henri Michaux with mescaline or Joan Mitchell with alcohol. There are also the Huichols from Mexico who created beautiful carpets inspired by their hallucinations during their pilgrimage to Wirikuta, during which they took peyotl to “communicate with the gods”. "Psychedelic art" developed in the late 1960s as a counterculture, featuring highly distorted or surreal visuals and bright colors, conveying psychedelic experiences. I am not so much attracted to this kind of art, preferring the rational/emotional spirit of Poussin or Matisse. My view is that authentic artists do not need drugs to be creative, even if a few of them may receive some incentive from them.

The works of art are artifacts, human productions, created for intersubjective communication. They use “symbolic forms” distinct from language, art is a mode of non-verbal communication of emotional states, imagination & rational experience, under the constraints of “rules of art”. Lev Vygotsky (1971) describes the art process as “emotional thinking”. Upon contemplation, the work of art shows an esthetic efficacy: an “explosive” union of emotions & imagination, referred to as *catharsis* by Vygotsky, which mobilizes conscious & non-conscious processes. In extreme cases, there is the so-called Stendhal effect. I quote Stendhal: “I was already in a sort of ecstasy ... absorbed in the contemplation of sublime beauty, I saw it up close, I touched it so to speak ... Leaving Santa Croce, I had a heartbeat ... life was exhausted in myself, I walked with the fear of falling”. Should we say that Stendhal describes some altered states of consciousness? In some cases, it may look like a pathological reaction of the brain, which needs medical help (a few cases per year at the Uffizi).

Most of the time, one may view the esthetic catharsis as a physiological phenomenon, a particular case of ignition. Ignition is an experimentally well-documented process that manifests itself through the sudden activation and invasion of the *global neuronal workspace* and monitors conscious access (Mashour et al., 2020). Thus, the hypothesis is that aesthetic experience mobilizes a particular type of ignition, an “esthetic ignition,” some kind of unforeseen global resonance reward within the GNW.

> "The hypothesis is that aesthetic experience mobilizes a particular type of ignition, an "esthetic ignition," some kind of unforeseen global resonance reward within the GNW."

The different components of the work of art (*e.g.*, in a painting) are brought together globally in the conscious workspace. The phenomenon is supramodal. It concerns innate dispositions — as in the recognition of shape, color, space, and movement, but also of figures, their dispositions, meanings, and emotional valence — epigenetically acquired through a long process of selection of synapses during the lifetime experience of the participant as stored long-term memory of events and emotions, personal history, education that was received, and the experience of other artists' works) (Changeux et al., 1973).

Esthetic ignition would be a naturally occurring physiological process — mobilizing jointly "reasons together with emotions" — that may occur in many cultures. Also, chills frequently accompanying a musical performance may be taken as a sign of esthetic ignition. Blood & Zatorre (2001) have recorded parallel increase/decrease of activation in defined brain territories, many of them being concerned by drug addiction. Does esthetic experience deserve the qualification of an "altered" state of consciousness? Why not? This is to be further discussed!

Generative neural networks have been increasingly used in artistic production. What is your opinion on the drawbacks and synergies that can operate between human and artificial intelligence when it comes to art and creativity?

Some of the AI so called "artistic productions" might be of interest, and even sometimes give some "catharsis". We are under conditions which resemble that of ChatGTP: you recover what you have used to feed the system, yet sometimes with unexpected organizations but often these are systematic *bric-à-brac* compositions of *clichés*. On the other hand the real human artist attempts to convey to the spectator/auditor the expression of his inner feelings, what one may call his "personality", and transmit his own personal message.

Art history enters into the actual style of the artist together with his own personal history, evolutive origins and the culture he/she belongs to. All that arises from the internalization in his brain of multiple experiences, education, unexpected encounters, original news… and of course the artistic context in which he/she is living. Nothing transcendental or irreducible in all that but the multiple intricate chances of epigenetic encounters, unpredictable chains of social circumstances, which make each individual human brain unique. There is no theoretical obstacle in principle with AI but machines simply miss an extraordinary intermingling between brain organization and cultural complexity, a theory of mind and individual history, among many other things which all are essential in artistic creation.

To end on a positive note, of course AI can be fruitfully used as a tool for creation. These are the plethora of new techniques that the artist may practice to express him/herself. These include, in music, computer assisted composition, the development of new musical instruments, of new categories of sounds but also of methods for treating and manipulating sounds on line. In visual arts, there is the management of new forms, the treatment of large numbers of data.

> “ Machines simply miss an extraordinary intermingling between brain organization and cultural complexity, a theory of mind and individual history, among many other things, which all are essential in artistic creation. ”

How do you perceive the role of artificial intelligence and machine learning in advancing our understanding of consciousness?

AI and machine learning offer new and very efficient tools for scientific inquiry, from molecular biology to ecology, of course, and up to brain and consciousness. Apart from this general practical issue, the question is: to what extent do authentic homologies exist in how the artificial machines and the human brain are constructed to process information and eventually generate consciousness?

There is a section of brain-inspired AI referred to as neuromorphic that attempts to develop silicon hardware resembling brain neurons and circuits. It has not yet developed and given results as outstanding as deep learning and ChatGTP with traditional AI. The principal aims of AI and machine learning are not to reproduce how the brain works but to simulate brain functions without necessarily any explicit reference to the brain. Spectacular successes have been achieved. However, there are intrinsic fundamental differences between machines and the brain that we cannot underestimate. Among them, AI machines benefit from a speed of processing orders of magnitudes faster (<speed of light) than the propagation of signals in the brain (<speed of sound). Also, they can embrace numbers of items with orders of magnitude larger than the human brain can do. Last, if the artificial simulation of conscious processing might have some successes in the future, silicons are not proteins and they shall miss (among many other features) the pharmacology of the real brain and the action of drugs… on consciousness!

In light of your groundbreaking work on the neural foundations of individual consciousness, how do you conceptualize the social dimension of consciousness?

Guillaume, you know that better than I do. I don't see a gap between individual and social consciousness since all of us know that the human brain is both rational and social. The social dimension has simply been missing in the past decades, and the time has come to reveal its importance. There is an urgent need for neuro-molecular conceptualization of social relationships such as theory of mind, decision-making, attachment, social bond, language, reasoning, science, art, ethics. The conceptualization should be directly inspired from the past work on individual consciousness yet with an original description of the relationships between individual consciousnesses. I have always been fascinated by the theories of Kropotkine (1902) on the importance of mutual aid in the competition between social groups, which according to him, are at the origins of *Homo sapiens* biological evolution. Yet the relationship between eventual genetic determinants of social life and biological evolution is still highly enigmatic.

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Considering this social dimension, does the idea of collective consciousness extend to suggest a form of consciousness that emerges from social groups or societies, akin to concepts of collective intelligence? What would be the implications for understanding societal behaviors and decision-making processes?

I have not been thinking about "collective consciousness" up to now. In any case, it should not be considered a metaphysical issue here. Since Durkheim (1912), it is usually considered as the set of shared beliefs, ideas, and moral attitudes uniting members within the society, creating solidarity, strengthening social bonds. There is, of course, an essential cultural component and, thus, a diversification of societies as a consequence of its epigenetic transmission. As a negative feature, it may lead to communitarianism and be the origin of multiple conflicts. When we see wars plaguing the world around us, we may realize that we are missing a collective consciousness that would share common *universal* values between conscious representatives of *Homo sapiens*. Would a stronger contribution of scientists improve the world situation? Not sure! But a science of collective consciousness should certainly develop!

What do you think are the most promising areas for future research in altered states of consciousness?

The most promising area is simply to further investigate, by similar methods, the neuro-molecular mechanisms of altered states of consciousness at the different levels or scales of brain physical organization (including the molecular level). This should include, in particular, the multiple levels of consciousness itself, such as minimal consciousness, recursive consciousness, self-consciousness, reflective consciousness, and theory of mind. These develop in the course of postnatal development in higher organisms, humans in particular, and in the course of biological evolution (see Lagercrantz & Changeux 2009, 2010).

I already mentioned the different modes of relationship with the environment *i.e.* body *vs.* outside world: physical, social, and cultural. Also, it is important to distinguish genetic from epigenetic conditions. All these are promising areas. The abundance of theories may be disorienting, and look useless, yet often they are complementary to each other and for all these reasons, there is the importance of open and unbiased scientific debate!

You have often advocated for interdisciplinary approaches in neuroscience. How do you think fields like computational psychiatry and social neuroscience can contribute to our understanding of altered states of consciousness? How do you see the role of international collaborations, like ALIUS, in shaping the future of consciousness research?

Althusser, in *Philosophie et philosophie spontanée des savants* (1967), criticized the slogan of "inter-disciplinarity" as leading to an internal fusion (or confusion) of disciplines that would then lose their identity. Personally, together with him, I prefer the term *multi*-disciplinarity. This means that representatives of different disciplines work together without losing the expertise they practice in their discipline, on the opposite. For instance, working on mental diseases like autism requires not only psychiatric expertise for the diagnosis but also human genetics, experimental psychology, brain imaging, developmental anatomy and chemistry, brain neurotransmitters and their receptors, synaptic epigenesis, at their most sophisticated levels. Nowadays, such expert multi-disciplinarity is needed to approach conscious processing successfully.

I may take this opportunity to emphasize the importance of the too-often-forgotten social sciences such as anthropology or ethnology. In this respect, the late Martin Fortier reached a remarkable level of multi-disciplinarity, uniting basic sciences and social sciences. In his studies of the use of hallucinogenic drugs in the Shamanic practices of native populations of South America. He was able to combine expertise in molecular pharmacology, behavioral sciences, and anthropology, all that in his single scientist brain. Yet, multidisciplinarity is most often achieved through collaborations between several experts from different labs working cooperatively on the same topic even if conflicts between "disciplinary egos" sometimes happen.

Collaborations are crucial to approach consciousness research, and international ones, such as ALIUS, are sometimes even more appropriate.

What advice would you give to young scientists embarking on research in the complex field of consciousness?

My first piece of advice to a young scientist is "to follow your own taste", as the poet Francis Ponge was saying, independently of fashion, financial pressure, scientific environment. As Andre Lwoff was saying: select a "good boss" who understands your intentions. Then, try to build a theoretical hypothesis — even simple-minded (better mathematical) — about your projects and develop the experimental tools to test it (see Volzhenin et al., 2022). In the field of consciousness, the efficient approaches are often multidisciplinary. Don't hesitate to learn about experimental techniques and theories in different labs — together with your own lab. This is a simple way to establish collaboration between labs. Avoid far-fetched theories and too complicated methods. Be simple minded and friendly. Good luck!

> "My first piece of advice to a young scientist is "to follow your own taste" independently of fashion, financial pressure, scientific environment."

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